

UNIVERSITY OF
Waterloo



Department of Mechanical and Mechatronics Engineering

Design of a Weatherproof Laptop for Rocketry Operations

A Report Prepared For:
The University of Waterloo

Prepared By:
Zachariah Mears
651 Monterey Ave
Victoria, BC, V8S 4T8

September 12, 2022

651 Monterey Ave
Victoria, BC, V8S 4T8

August 1, 2022

Prof Derek Wright
Director, Mechatronics Engineering
Department of Mechanical and Mechatronics Engineering
University of Waterloo
200 University Ave W.
Waterloo, Ontario, N2L 3G1

Dear Professor Wright,

This report, Design of a Weatherproof Laptop for Rocketry Operations, was written to fulfil my 4A work term report requirements. This is my 3rd work report submission.

This report covers the requirements and problem definition, along with the early design work, for a new weatherproof laptop for the rocketry team. The concept is a large laptop with 3 displays, that is rugged and impervious to any water or dust, as well as being resistant to the very wide range of temperatures that the team conducts operations in. It will be used at almost every team event, as well as for normal work in the team's design bay, and as such will provide a very large amount of value to the team.

This report was written entirely by me and has not received any previous academic credit at this or any other institution. I would like to acknowledge the help provided by the members of the Waterloo Rocketry team, who provided input and feedback on the project requirements and initial design, as well as allocating the funding that will be required to finish building the project. I would also like to thank Framework Computer Inc for supporting the project by providing much of the needed hardware at no cost as part of their developer program.

Best Regards

Zachariah Mears
ID: 20792815
4A Mechatronics Engineering

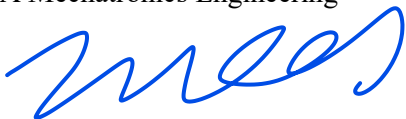


Table of Contents

List of Figures	2
List of Tables	3
Summary	4
1.0 Introduction.....	5
1.1 Current Setup	5
1.2 Proposed Setup.....	6
1.3 Industry Support.....	6
1.4 Objective	6
2.0 Problem Definition.....	7
2.1 Requirements	7
2.2 Criteria	10
3.0 Design	11
3.1 Overview.....	11
3.2 Displays.....	12
3.3 Inputs and Outputs	13
3.4 Battery.....	13
3.5 Frame	14
3.6 Remaining Challenges	16
4.0 Conclusions.....	17
4.1 Conclusions.....	17
4.2 Next Steps	17
References.....	19

List of Figures

Figure 1: Framework Mainboard.	11
Figure 2: System Block Diagram.	12
Figure 3: Framework display panel.	13
Figure 4: Front view of preliminary OpsTop design.	14
Figure 5: Back view of preliminary OpsTop design.	15

List of Tables

Table 1: OpsTop System Requirements.....	7
Table 2: OpsTop Project Criteria.	10

Summary

Waterloo Rocketry is a design team from the University of Waterloo that focuses on designing and launching high powered sounding rockets [1]. Throughout the year, particularly during the winter, the team conducts a variety of outdoor tests to develop and characterize the team's rocket engine. Each June the team then travels down to the New Mexico desert to compete in the Spaceport America cup and launch the rocket. Between testing in the Canadian winter and flying in the middle of summer in the desert, the team must run operations in a very wide range of extreme environmental conditions, including dust, rain, snow, and very cold and hot temperatures. As such the team designs its equipment to handle such extreme conditions. However, over the past few years the team has developed new data management software which requires the use of modern computers for these operations. Currently the team relies on personal consumer laptops, which are not designed to handle or operate effectively in these conditions. In addition to poor performance, this introduces an unacceptable amount of risk to team member's personal property.

To address this problem, a new ruggedized laptop, known as the Operations Laptop (OpsTop) is being developed to meet the teams requirements. The team has partnered with Framework Computer Inc who is providing much of the necessary hardware in the form of parts from the Framework laptop. With this support. A preliminary design has been created which should meet or exceed the team's requirements.

The OpsTop is based around the mainboard of the Framework laptop, which features a powerful Intel i5 processor [2], along with sufficient memory and storage to make it a very capable machine. It will feature three displays for plotting large amounts of data, along with a very large battery for long run times, and two modular payload bays, allowing for various operation specific hardware, such as radio systems to be installed.

At present, the preliminary design phase has been completed, and most of the key design features of the laptop have been determined. This includes the frame design and construction techniques, water and dust sealing methods, and the system architecture, including all major connections between components.

The team is currently waiting to receive many of the key components from Framework once these are received the project will proceed with detailed and final design of all mechanical components, along with electrical integration testing and custom battery development. The goal is to get the system running by the end of 2022 such that it can be used for all rocket engine tests and other operations starting in the new year.

1.0 Introduction

Waterloo Rocketry is a student design team at the University of Waterloo [1]. The team builds high powered sounding rockets with a goal of reaching a peak altitude of 30,000ft. Each June the team travels down to the New Mexico desert to compete in the Spaceport America Cup. In addition to flying the rocket at competition, the team conducts a variety of engine tests throughout the year, particularly during the winter in Waterloo.

The result of launching during the summer in the desert and conducting testing outside in the Canadian winter is that all the equipment used by the team needs to be capable of withstanding a wide range of environmental conditions. Specifically, equipment needs to handle the extreme heat of the desert, and cold of the test site. In addition, it needs to handle all forms of precipitation from sudden monsoons in New Mexico, as well as Blizzards during the winter.

Because of this requirement for extreme weather resistance, all the custom launch control and data acquisition systems designed by the team are built to be weatherproof units, using waterproof pelican cases with weather resistant connectors. However, the team has encountered issues when off the shelf systems need to be used. Particularly, over the past two years, the team has developed software for adaptable transfer, plotting, and logging of data. Known as Omnibus [3], this system has proven incredibly powerful for effectively running tests and launches. However, it requires the use of full capable computers and networking equipment to run.

1.1 Current Setup

Currently the team owns a single Lenovo ThinkPad. This is used at the launch/test site to host the Omnibus server and collect data from the data acquisition system. The data is sent to mission control via a pair of outdoor long range Wi-Fi antennas to mission control, where it is displayed on personal laptops belonging to the operations team. While the ThinkPad was originally selected due to having impressive durability for a consumer device, it is not weatherproof, and requires significant time be spent each test/launch to ensure that it is protected from the elements. While the make and model of the personal laptops used at mission control varies, they are all equal or worse durability compared to the team's ThinkPad. This is exacerbated by the fact that since they do not belong to the team, any potential damage to them is entirely unacceptable.

1.2 Proposed Setup

To improve the robustness and ease of setup of the Waterloo Rocketry ground infrastructure, the goal is to remove the need for any consumer grade laptops from the setup. At the launch tower / test site a small single board PC will be placed inside the waterproof case of the data acquisition system. This is a simple and effective solution as there is no need for a user interface on this computer.

At mission control, a fully capable laptop is still required for setup, debugging, and most importantly plotting of large amounts of data. Currently this large amount of data necessitates the use of multiple laptops so that there are more screens to view the data on. Commercial off the shelf rugged weatherproof laptops are available, and there are even options intended for military use which feature multiple displays that would allow the use of a single machine to plot all the data. Unfortunately, these laptops are prohibitively expensive given the team's limited budget. As such it has been decided that the best solution is for the team to develop our own ruggedized laptop to perfectly suit the team's needs.

1.3 Industry Support

In April of 2022, Framework Computer Inc, a laptop startup focused on repairability and sustainability, announced a contest where they would give away 100 mainboards for use in interesting projects. These mainboards feature an 11th generation Intel i5 processor [2], and with the addition of an SSD and RAM, act as a powerful windows or Linux single board computers. The idea to use one of these mainboards as the heart of the team's new ruggedized laptop was submitted, and ultimately was selected. As such the team received the mainboard for free, eliminating the need to purchase the most expensive component of the laptop, and further increasing the incentive for the team to develop its own laptop. Recently, Framework has also agreed to further support the project by providing the displays, storage, and other necessary laptop parts.

1.4 Objective

Waterloo Rocketry routinely conducts operations in a wide range of extreme weather conditions. As part of these operations, the team requires a fully capable windows laptop for plotting test and launch data and debugging any issues. Due to the lack of affordable off the shelf options, there exists a need for the creation of a custom ruggedized and weatherproof laptop for use in these operations.

2.0 Problem Definition

As stated, the goal of this project is to build a ruggedized laptop which can be used for plotting system data and running operations. The system will be known as the Operations Laptop (OpsTop). This system needs to handle the wide range of extreme weather that the team operates in. It must also have space for the radio modules used for communicating with the rocket and have multiple displays for allowing multiple operators to review large amounts of data. The following section provides the detailed set of requirements that have been generated based on the team's needs.

2.1 Requirements

The requirements for this project are based on the team's experience using a normal laptop for operations for the past two years, and the issues that have been observed when doing so. Table 1 shows this set of requirements, which has been laid out and reviewed by the team and defines the necessary capabilities of the OpsTop.

Table 1: OpsTop System Requirements.

Number	Name	Description	Justification
1. Environmental			
1.1	Temperature	The laptop must be designed to handle temperatures from -30 degrees to +40 degrees Celsius, in direct sunlight.	Maximum range of temperatures the team has observed during operations
1.2	Weather	The laptop must be able to handle heavy rain and snow, when either open or closed, and must withstand high winds while closed, or open with support.	Range of weather conditions the team has operated in in the past
1.3	Puddles	The laptop should withstand being placed upright in puddles up to 20mm in depth.	Maximum depth of puddle an operator is likely to try and work in

1.4	Sunlight	The laptop must be reasonably usable in direct sunlight for short periods of time, and comfortable to use for extended periods when in light shade.	The laptop will be used for quick debugging out in the open, but mostly used in the shade for operations
1.5	Dust	The laptop must be completely resistant to dust. Fans and filters may have only limited protection but must be easily replaced in the field if prone to damage.	The desert is dusty
2. Physical			
2.1	Weight / Portability	The laptop should be light enough that it can comfortably be carried for at least 15 minutes when closed by any team member.	15 minutes is the longest walk that is likely before a vehicle is used
2.2	Size	The laptop should be small enough to be moved by any single team member reasonably easily when either open or closed.	Operators need to be able to work independently
2.3	Durability	The laptop should be built to be very durable; it should survive being dropped out of the bed of a truck and then dragged 100ft across gravel.	Systems tend to endure large amounts of accidental abuse
2.4	Ergonomics	The laptop should be designed to be reasonably comfortable to use for two hours, such that it can also be used as a regular computer for programming, talking on messaging apps such as slack, and more as required.	The laptop will also be used throughout the year as a communal workstation.
2.5	Labelling	All Connectors and switches should be clearly and professionally labelled using a method that will withstand the required abuse and conditions as listed above.	Some operators will have no experience when using the system, so it must be fool-proof

3. Performance			
3.1	Computer Specs	The laptop must have plenty of processing power, RAM, and solid-state storage, such that it can easily run as a fully capable windows machine. (Recommended: i5 11th gen, 16GB of RAM, 500GB SSD or better)	There are a number of software tools that need to be used which require a powerful modern machine
3.2	Monitors	The laptop must have three, built in displays, each with a resolution of at least 1080p.	There is a large amount of data that needs to be displayed
3.3	Battery	The laptop should have a battery life of 12h and be able to reach full charge in 5h when not running to allow short overnight charging.	The laptop is planned to be used in the desert without access to charging outlets, and must be charged during short nights at hotels
4. Connectivity			
4.1	Networking	The laptop must have a standard Wi-Fi card, and an accessible ethernet port.	The team uses both ethernet and Wi-Fi for operations
4.2	Human Interfaces	The laptop must feature a full-sized keyboard, and some combination of a mouse, trackpad, drawing tablet, or similar, with consideration put into usage while wearing gloves. It must also feature an aux jack to allow online calls, speakers, a mic, and a webcam are desirable.	The laptop will be used on a regular basis as a normal workstation
4.3	Payload Bays	The laptop should feature two modular payload bays that provide power and data. The first of these will be used for radios for interfacing with the rocket, with the second reserved for future expansion.	The team needs to communicate with the rocket and there are

			plans for other future systems
4.4	External IO	In addition to previously listed connectivity, and any internal connections to payload bays, the laptop should feature one external thunderbolt port and two USB A ports	Other peripherals will need to be connected to the laptop in the future
4.4	Power	The laptop should be compatible with the standardised GSE connector for power / charging, as well as having an AC charger.	The laptop will be charged either using the team's custom battery packs, or standard wall power

2.2 Criteria

In addition to the listed requirements, there are several criteria which apply to the OpsTop. The goal of these criteria is to ensure the project remains affordable, ensure it provides valuable learning opportunities to team members, and to ensure that it is completed quickly. Since the system will provide immense value to the team's ability to conduct effective operations, it is important that the system be operational as soon as possible. Table 2 lists these criteria, along with descriptions and justifications.

Table 2: OpsTop Project Criteria

Number	Name	Description	Justification
1	Cost	The cost of the project should be minimized and should not exceed 1,500\$.	This is the maximum amount the team has allocated for the project
2	Manufacturing	The system must be made using facilities available on campus and primarily done by team members	The goal of the team is to promote learning and so whenever possible projects should be used to teach members
3	Timeline	The project must be completed by the end of 2022	The team will be running many poor weather operations over the

			coming year, and this system is needed as soon as possible
--	--	--	--

3.0 Design

3.1 Overview

The core of the OpsTop will be built around a Framework laptop mainboard running windows. This mainboard has four USB-C Thunderbolt ports, as well as connectors for a display, trackpad, and battery [2]. Figure 1 shows an example mainboard identical to the one that will be used in the OpsTop.



Figure 1: Framework Mainboard.

The mainboard will connect to all the other major components of the laptop, including the displays, large custom battery, hub for connecting to payloads, and trackpad. The mainboard is also where the OpsTop's SSD, RAM, and Wi-Fi card mount. Framework is providing 16GB of ram and a 512GB SSD which will allow the system to meet the team's performance requirements (requirement 3.1). They are also providing a standard wifi card which combined with a USB hub for ethernet, ensures the system will meet the team's networking needs (requirement 4.1).

A block diagram of the entire OpsTop is shown in figure 2. This diagram shows all of the major connections internal connections in the laptop. It is however missing some smaller details such as fan connections, and connections between the main USB hub and the outer case of the laptop.

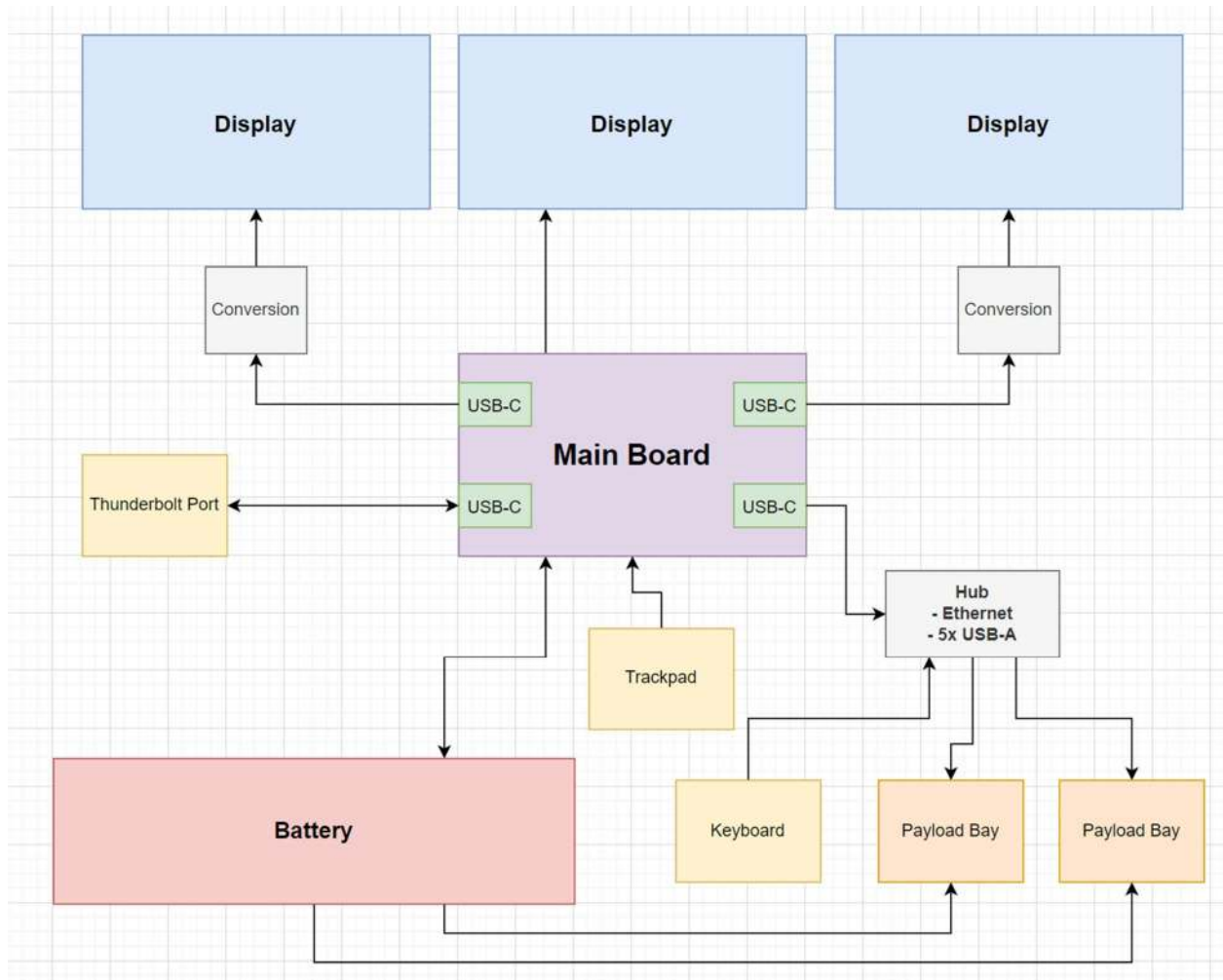


Figure 2: System Block Diagram.

3.2 Displays

The OpsTop will use 3 Framework laptop displays for the monitors (requirement 3.2). The Framework laptop displays were selected since they will be provided for free by Framework and provide good outdoor performance. They are also inherently compatible with the framework mainboard. Since the mainboard only has a connector for one display, adapters are needed for the other two displays. This consists of a pair of USB-C to display port adapter cables, which will feed into display port to eDP adapter boards allowing them to connect to the eDP connector on the displays. An example of one of the three display panels that will be used is shown in figure 3.



Figure 3: Framework display panel.

3.3 Inputs and Outputs

The laptop will feature a standard keyboard and trackpad for navigation and typing (requirement 4.2). For the keyboard, a silicone waterproof USB keyboard will be used. This type of keyboard was selected as they are cheap and fully waterproof, making them easy to integrate into the system. For the trackpad, a standard Framework laptop trackpad will be used. To make the trackpad waterproof an adhesive film will be used to cover it and the surrounding frame. This ensures a watertight seal between the two, while still allowing the trackpad to be used normally. While it is not necessary for the initial completion of the laptop, it is intended that a waterproof Bluetooth mouse will also be sourced for use with the OpsTop. This will allow it to be used more easily while an operator is wearing gloves during the winter.

3.4 Battery

Since the laptop features three displays, which will usually be run at full brightness and several other peripherals, it will draw more power than a typical Framework laptop. The Framework laptop typically has a battery life of about 4 hours. Since the OpsTop needs a battery life three times that of a typical Framework laptop (requirement 3.3) and the OpsTop is expected to draw almost twice as much power, it will need a much larger battery.

The battery will feature a custom battery management system and controller built by the team. The pack itself will be built out of a total of 32 1860 lithium-ion battery cells. This gives it a total capacity of 260Wh, or five times that of a normal framework laptop. The cells will be arranged into a 4S8P configuration, giving it the same voltage as the stock battery and allowing it to interface with the

mainboard just like the stock battery. To ensure this large battery pack is built safely and effectively the team will consult with other design teams in the Sedra Design Center, many of which have extensive experience building similar battery packs for electric vehicles.

3.5 Frame

The body of the laptop will be made from sheet aluminum. The pieces will be waterjet cut to ensure accuracy before they are bent and welded to form the base of the frame. The top cover of the base will be made from polycarbonate to allow visual inspection of the inside of the machine, as well as to allow the Wi-Fi and Bluetooth signals to penetrate the case.

The lid of the laptop will be made from a thicker single piece of aluminum. A CNC router will be used to cut out pockets for the displays and other mounting geometry. A piece of polycarbonate will cover the displays to protect them.

The use of aluminum and polycarbonate will ensure that the case is very durable (requirement 2.3) while keeping the weight of the system relatively low (requirement 2.1). Figure 4 shows the front view of the current preliminary design. The aluminum frame and lid can be seen in grey, while the polycarbonate appears as the transparent material covering the internals and displays of the laptop.

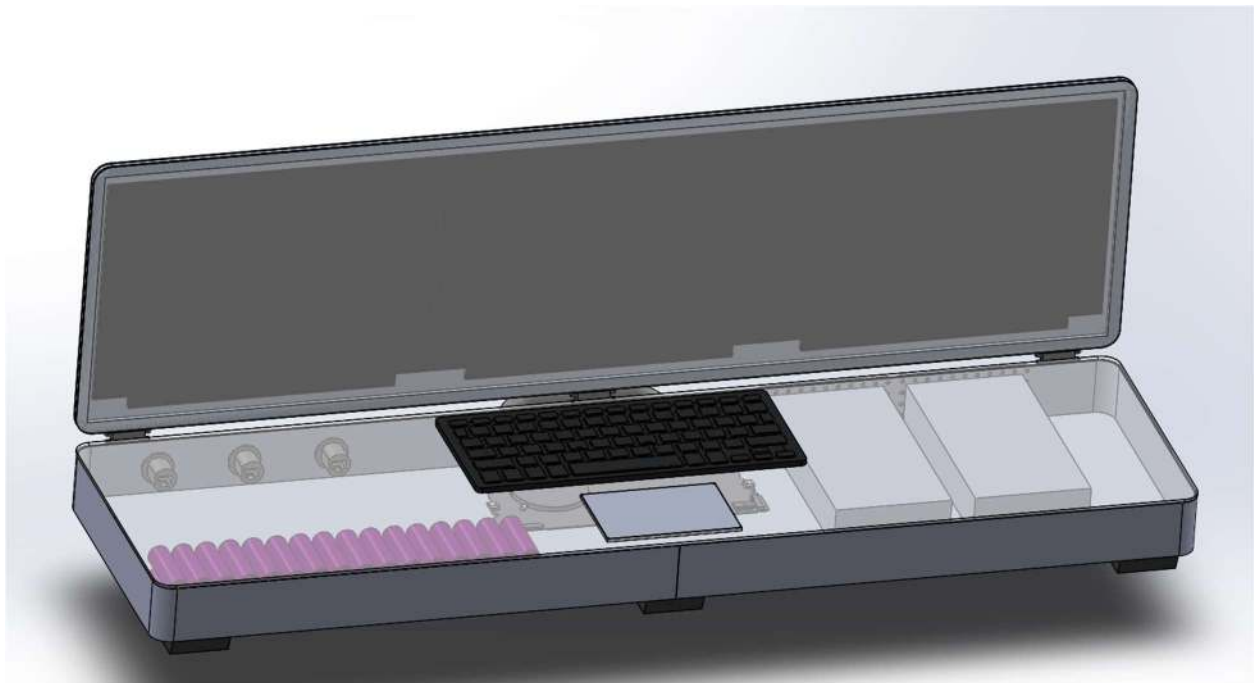


Figure 4: Front view of preliminary OpsTop design.

To prevent water from entering between the polycarbonate and the aluminum case and lid, as well as where the payload bays attach into the back of the laptop, rubber seals will be used. They will either be laser cut or waterjet cut depending on the exact material selection, and a tightly spaced bolt pattern will be used to ensure a good seal. The seal geometry is missing from the above figure, but in the lower left-hand corner of figure 5 the bolt pattern used for sealing the payload bays to the case can be seen. To seal the display and USB keyboard cables watertight panel gromets will be used.

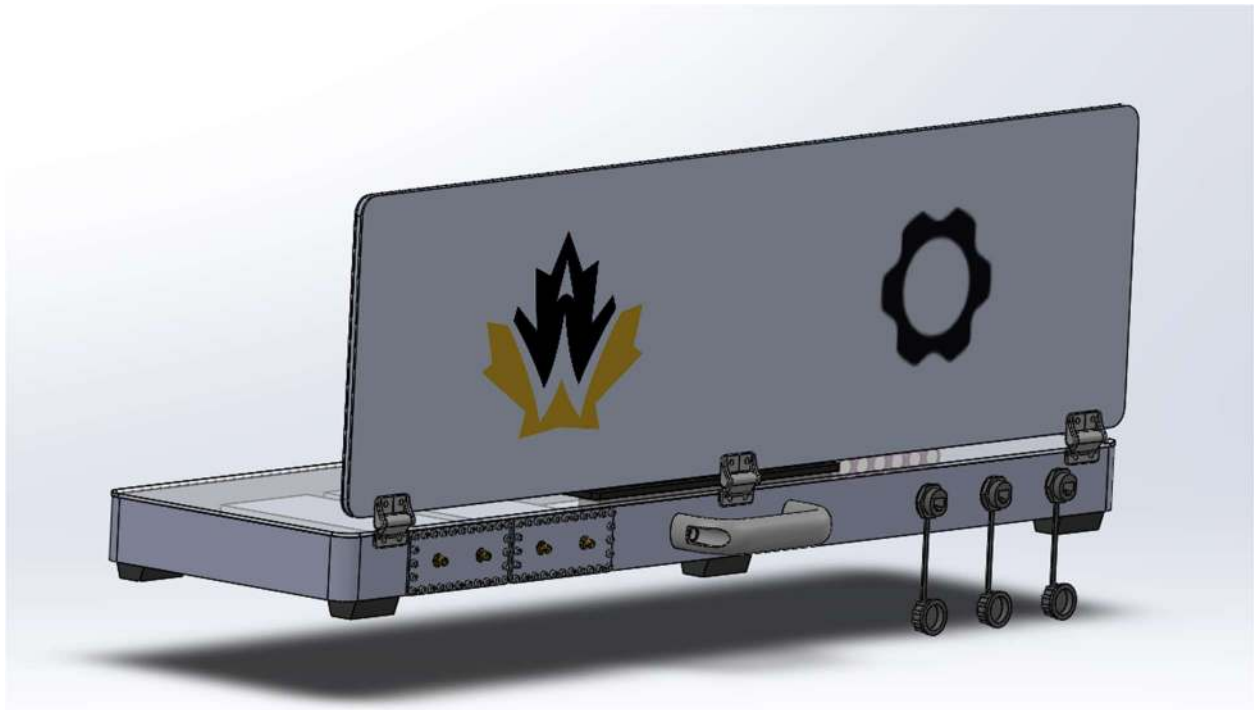


Figure 5: Back view of preliminary OpsTop design.

For mounting the lid to the base three constant friction hinges, McMaster-Carr PN 2190A23 [4], will be used. These hinges provide a constant amount of resistance to motion, allowing the laptop lid to be adjusted then hold its position, just like any standard consumer laptop. To keep the laptop closed during transport magnets will be used in the corner of the display and base, allowing it to snap shut similar to consumer devices.

To add grip when the system is used on smooth tables, and to allow some clearance from wet ground when used in the field, the OpsTop will use a total of six feet mounted on the bottom. At 15mm tall these feet will be much taller than those normally found on laptops. They will be 3D printed in a soft rubber like material for good grip and shock absorbance and will be mounted via bolts into the base of the

frame. The last major piece of hardware on the exterior of the case is the handle. This is a simple large aluminum handle, McMaster-Carr PN 5190A31 [5], which will allow the laptop to be comfortably carried to different areas of the test or launch site (requirement 2.1).

While the aluminum and polycarbonate frame will provide a very hard and robust case. Without additional internal supports, it is likely that it will be too flexible. In addition to this, the aluminum case will not provide good mounting points to directly attach the internals of the laptop, such as the battery and mainboard. Finally, the case will require substantial effort and cost to manufacture, and so cannot be easily modified or re-made if the team wishes to modify the laptop internals. To resolve these issues, an internal frame made from 3D printed PETG components will be used. These parts will bolt into the aluminum frame using a standard mounting pattern. They will serve to add rigidity to the frame and provide secure mounting points for all of the internal components of the laptop. If any internal parts of the laptop need to be modified, then the cheap plastic parts can simply be re-printed, without needing to modify the outer case.

3.6 Remaining Challenges

As the design of the OpsTop is still in the preliminary phase, there are a number of challenges which remain to be solved. One of the largest remaining problems is thermal management of the laptop. The Intel i5 processor will generate up to 28W of heat when running [6]. In a typical laptop, fans can easily be used to blow cool outside air over a heatsink to keep the processor cool. Since the OpsTop needs to be watertight, this cannot be done directly. In addition, most consumer laptops are not designed to be continuously used in 40 degree heat, and the much hotter ambient air means that the cooling becomes less effective. Currently the intention is to use heat pipes to transfer heat from the processor through a watertight bulkhead in the case to a heatsink and powerful waterproof fan. This will allow dusty or wet air to still be used for cooling without risking damage to the processor. By using a much larger fan and heatsink than is typically used it is hoped that this can make up for the less effective cooling due to the warmer ambient air. There remains a great deal of design work to ensure this approach will be effective and practical to implement.

While a concept for the design of the watertight seals has been created, this approach still needs to be verified. The team also needs to develop a small watertight network switch, and the intention is to use the same construction techniques for this much smaller project as those that will be used on the OpsTop in order to verify that the seal design works as intended.

There is also still a large amount of detailed design that must be completed before the final system can be built. This includes the mechanical design of the internal frame, sealing flanges, and detailed design of the lid. There is also a large amount of electrical design that needs to be completed for the custom battery pack, as well as the team may need to build their own Display Port to eDP adapter boards depending on the performance of the available off the shelf systems.

4.0 Conclusions

4.1 Conclusions

The Waterloo Rocketry team requires a rugged and powerful laptop for running their operations in all weather conditions. The team's current Thinkpad laptop, as well as the personal laptops that are borrowed for these operations are not sufficient for the task, and the current risk of damage to personal property is unacceptable. As such, the team has defined a set of requirements for a new system that will be robust enough to easily handle the extreme conditions within which the team operates, while providing significantly better performance for the needs of the operations team. Thanks to the donation of parts from Framework Computer Inc, the team will be able to build such a system at a reasonably low cost. A preliminary design has been created which should meet or exceed all of the requirements which have been set out. This preliminary design has been discussed and reviewed, and the project is now going to begin on the detailed design and construction of the system.

4.2 Next Steps

The team is currently waiting on many of the key components to arrive from Framework. Once these parts have been received the project will be able to rapidly progress with prototyping and development. One of the first major tasks will be to finish sourcing all of the minor components of the laptop and creating CAD models of all of the parts. From there the detailed CAD of the entire system can be completed.

Simultaneously, the electronics will be laid out on a flat board so that the integration and functionality of all the components can be tested before final integration of the system. This setup will also be used to develop the necessary battery management circuitry, and to test the custom battery pack. With the electronics working, testing and measurements can be done to finalize the design of the thermal management system.

Once the electronics and thermals have finished being developed and tested, and the mechanical design has been finalized, there will be a detailed design review of the system. This will allow team members to provide feedback on the design, and to find any issues with the design. Once any outcomes from the design review have been addressed, manufacturing of the OpsTop can begin. Once the frame is manufactured, the system can be integrated into its final form. The final step for the laptop will be its commissioning and use at the next rocket engine test, where its performance for running operations can be evaluated.

References

- [1] Waterloo Rocketry. (n.d.). *About Us*. Home - Waterloo Rocketry. Retrieved September 16, 2022, from <https://www.waterloorocketry.com/>
- [2] Framework Computer Inc. (n.d.). *Mainboard*. Framework. Retrieved September 16, 2022, from <https://frame.work/ca/en/products/mainboard?v=FRANFG000A>
- [3] Waterloo-Rocketry. (n.d.). *Waterloo-Rocketry/Omnibus: It's a bird... it's a plane.. it's omnibus!* GitHub. Retrieved September 16, 2022, from <https://github.com/waterloo-rocketry/omnibus>
- [4] McMaster-Carr. (n.d.). *Friction Hinge*. McMaster. Retrieved September 16, 2022, from <https://www.mcmaster.com/2190A23/>
- [5] McMaster-Carr. (n.d.). *Oval Pull Handle*. McMaster. Retrieved September 16, 2022, from <https://www.mcmaster.com/5190A31/>
- [6] Intel Corporation. (n.d.). *Product specifications*. Intel Core i51135G7 Processor 8M Cache up to 4.20 GHz Product Specifications. Retrieved September 16, 2022, from <https://ark.intel.com/content/www/us/en/ark/products/208658/intel-core-i51135g7-processor-8m-cache-up-to-4-20-ghz.html>